Israel Souza Ferreira

Data Scientist | Machine Learning

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PROFILE

Data scientist working across the full problem cycle: extracting and preparing data at scale, statistical and predictive modeling, experimental design, and communicating results. R&D experience as a CNPq research fellow applying **Item Response Theory**, **time-series forecasting**, **language models**, and rigorous model evaluation to problems in education and public health. Computer Science background with solid grounding in statistics, optimization, and SQL. What sets me apart is evaluation: measure before concluding, and distrust your own result.

PROFESSIONAL EXPERIENCE

Tieta Artificial Intelligence — **Remote, Brazil**
Data Scientist / R&D Researcher — CNPq RHAE Research Fellow — *Apr 2025 - Jun 2026*

- Modeled latent psychometric parameters of educational items with **Item Response Theory**, validating model estimates against reference parameters.
- Designed **paired evaluations** over the same items to compare competing approaches under the same protocol, quantifying both the correlation of each estimated signal with reference parameters and its inference cost.
- Built signal-extraction pipelines with **language models** orchestrated by **DSPy**.
- Tested **multitask regression** architectures over textual and visual data.
- Structured reproducible experiments, evaluation metrics, and technical documentation to support the conclusions of the research cycle.

Tieta Artificial Intelligence — **Remote, Brazil**
R&D Researcher — Research and Innovation Training Program — *Jul 2026 - Present*

- Developing a **question segmentation** pipeline for scanned exam documents via OCR, extracting structured items for educational analysis.

Independent Consulting — Educational Institution — **Remote, Brazil**
Full Stack Developer and Machine Learning Engineer — *2026 - Present*

- Delivered a contracted end-to-end facial recognition attendance system, functional and in final acceptance testing before deployment, covering the interface, modeling, data, backend, testing, and deployment.
- Trained an EfficientNet-B0 classifier for automated medical-document triage using **hyperparameter search** (Optuna), stratified cross-validation, and a hybrid CNN + OCR decision pipeline; diagnosed underfitting through **controlled ablation** and raised validation accuracy by **13 percentage points**.
- Implemented 1:N **vector search** and 1:1 verification over 512-dimensional facial embeddings with InsightFace and pgvector, aggregating *multi-frame* decisions to reduce false positives, with *liveness* detection and anti-replay protection.
- Built the data layer and inference service with PostgreSQL, SQLAlchemy, Alembic, and FastAPI; delivered Dockerized infrastructure with HTTPS, GitHub Actions, and **315 automated tests**.
- Resized the delivery to the client's actual server — no GPU, 8 GB of RAM — cutting **8 GB** from the images and **2 GB** from memory usage by moving inference to **ONNX Runtime**, with numerical parity verified against PyTorch (bit-identical preprocessing, **zero** decisions changed). Validating the package by installing it from scratch surfaced **three defects** that would have blocked installation and were out of reach for unit tests.

Federal University of Ceará — Quixadá Campus — **On-site, Brazil**
Teaching Assistant for Algorithms, Calculus, and Pre-Calculus — *2022 - 2024*

- Provided academic support in data structures, algorithms, and complexity analysis, translating quantitative material for undergraduate classes.

SELECTED PROJECTS

- **Dengue Case Forecasting (time series):** forecasting case notifications from Brazilian SINAN surveillance microdata, with ETL, LSTM and PLE models, experiment tracking in MLflow, and an inference service; reduced **RMSE from ≈98 to 83** on the Ceará series (≈15% drop). PostgreSQL, PyTorch, FastAPI, and Docker.
- **Brazilian Emergency Aid — Performance Analysis over 31.6 GB:** university assignment picked back up on my own initiative and rebuilt for real scale over **31.6 GB** of public data; per-stage instrumentation correlated memory growth with retained deduplication identifiers (**r = 0.99**), isolating the root cause of the bottleneck. Chunked ingestion through asynchronous binary `COPY`, PostgreSQL, Pandas, and Streamlit.
- **Triple Roman Domination in Graphs (undergraduate thesis):** first metaheuristic approaches reported in the literature for the problem, using a Genetic Algorithm, MAX-MIN Ant System, and RVNS, with Integer Linear Programming as an exact validation *baseline*.

TECHNICAL SKILLS

Languages and querying: Python, SQL, C++, and Bash.

Analysis and statistics: Pandas, NumPy, SciPy, Scikit-learn, hypothesis testing, correlation analysis, cross-validation, model calibration, ablation analysis, and Item Response Theory.

Modeling and machine learning: PyTorch, Optuna, DSPy, FAISS, InsightFace, OpenCV; time series (LSTM, PLE), multitask and multimodal regression, MoE/MMoE, attention, vector search, and metaheuristics.

Data, MLOps, and communication: PostgreSQL, pgvector, SQLAlchemy, Alembic, MLflow, Docker, Git, GitHub Actions, GitLab CI/CD, FastAPI, pytest, mypy, Streamlit, data visualization, and technical documentation.

EDUCATION

Federal University of Ceará — Quixadá Campus — **Brazil**
B.Sc. in Computer Science — *Jul 2026*

LANGUAGES

Portuguese: Native | **English:** Fluent technical reading of research papers and documentation